

[SINGLE CORRECT CHOICE TYPE]

Q.1 to Q.7 has four choices (A), (B), (C), (D) out of which **ONLY ONE** is correct.

Q.1 If $f(x) = \min. \{ |x-1|, |x-2|, |x-3| \}$, then the value of $I = \int_0^3 f(x) dx$ equals

- (A) 1 (B) 2 (C) 3 (D) 4

Q.2 Let $f(x) = \frac{x}{x-1}$ and $g(x) = \underbrace{(f \circ f \circ \dots \circ f)}_{f \text{ occurs } 2013 \text{ times}}(x)$, then $\int_{1/6}^{1/3} g(x) dx$ equals

- (A) $\ln \left(e^{\frac{1}{2}} \cdot \frac{4}{5} \right)$ (B) $\ln \left(e^{\frac{1}{6}} \cdot \frac{4}{5} \right)$ (C) $\ln \left(e \cdot \frac{4}{5} \right)$ (D) $\ln \frac{4}{5}$

Q.3 Let $f(x) = \frac{4x^2 - 3x + 1}{2\pi x}$. If $f' \left(\frac{1}{2} \right) = \frac{p}{q\pi}$, $p, q \in \mathbb{N}$, then $(p-q)$ equal

- (A) 6 (B) 5 (C) 4 (D) 3

Q.4 If $\int e^{1-\cos x} (1+x \sin x) dx = f(x) \cdot e^{1-g(x)} + c$, then number of solutions of the equation $|f(x)| = g(x)$ equals (c is arbitrary constant)

- (A) 0 (B) 1 (C) 2 (D) 3

Q.5 If $A = \{a_1, a_2, \dots, a_n\}$ be set of all possible natural factors of 1000 and $P = \int_{-2}^2 x \ln(a_1^x + a_2^x + \dots + a_n^x) dx$,

then value of $\frac{2P}{n}$ equals

- (A) $\ln 10$ (B) $\ln 2$ (C) $\frac{\ln 2}{16}$ (D) $8 \ln 10$

Q.6 Consider a real valued continuous function f such that $f(x) = \sin x + \int_{-\pi/2}^{\pi/2} (\sin x + t f(t)) dt$.

If maximum value of the function is equal to $k(\pi+1)$, $k \in \mathbb{N}$, then the value of k is

- (A) 1 (B) 2 (C) 3 (D) 4

- Q.7 $\int \frac{x - \sin x \cos x}{x^2 \cos^2 x} dx = F(x)$ such that $F(1) = 1$, then $\lim_{x \rightarrow 0} F(x)$ equals
- (A) 1 (B) $1 - \tan 1$ (C) $2 - \tan 1$ (D) $\tan 1$

[PARAGRAPH TYPE]

Q.8 to Q.10 has four choices (A), (B), (C), (D) out of which **ONLY ONE** is correct.

Paragraph for question nos. 8 to 10

Let a continuous function f satisfies the relation

$$(1 + x^2) f(x) \cdot f\left(\frac{2}{x}\right) + x = x f(x) + (1 + x^2) f\left(\frac{2}{x}\right) \quad \forall x \in \mathbb{R} - \{0\} \text{ and } f(0) \neq 1$$

- Q.8 If $\lim_{x \rightarrow 0} (\tan^{-1}(f(x)) + a)^{\frac{4}{x}} = e^p$, then p is equal to
- (A) 1 (B) 2 (C) 4 (D) 8
- Q.9 If $\int_0^{\infty} \frac{\ln(2f(x))}{4+x^2} dx = \frac{-\pi \ln 2}{\lambda}$, then λ is equal to
- (A) 1 (B) 2 (C) 4 (D) 8
- Q.10 If $g: [0, 2] \rightarrow \left[0, \frac{1}{2}\right]$ and $g(x) = f(x)$, then $\int_0^{1/2} g^{-1}(x) dx$ is equal to
- (A) $1 + \ln 2$ (B) $2 - \ln 2$ (C) $1 + 2 \ln 2$ (D) $1 - \ln 2$

[MULTIPLE CORRECT CHOICE TYPE]

Q.11 to Q.14 has four choices (A), (B), (C), (D) out of which **ONE OR MORE** may be correct.

- Q.11 If $I_1 = \int_0^{\pi} \frac{\cos x}{(x+2)^2} dx$ and $I_2 = \int_0^{\pi/2} \frac{\cos x \cdot \sin x}{(x+1)} dx$ and $\lambda I_2 = \frac{2}{\mu + \pi} + k - \gamma I_1$, then
- (A) $\lambda = 4$ (B) $\mu = 2$ (C) $k = 1$ (D) $\gamma = 2$
- Q.12 If $f(x) = \int_0^x (f(t))^{-1} dt$ and $\int_0^1 (f(t))^{-1} dt = \sqrt{2}$, then
- (A) $f(2) = 2$ (B) $f(8) = 4$ (C) $f(8) = 9$ (D) $f(128) = 8$

Q.13 If $f(x) = \sqrt{1 - \frac{1}{x}}$ and $g(x) = \sec^{-1} x$ then identify which of the following statement(s) is(are) correct

(A) Range of $g(f(x))$ is $\left(0, \frac{\pi}{4}\right]$ (B) Domain of $g(f(x))$ is $(-\infty, 0) \cup [1, \infty)$

(C) $\int_{-4}^{-1} \cot(g(f(x))) dx = \frac{14}{3}$ (D) $\lim_{x \rightarrow -\infty} \left(\frac{\cot(g(f(x)))}{\sqrt{1-x} + \sqrt{2-x}} \right) = \frac{1}{2}$

Q.14 A function f satisfies $2 + \int_2^x f(t) dt = \frac{x^2}{2} + \int_x^2 t^2 f(t) dt$, then number of real solutions of the equation

$x^2 - 2x + 2(1 - f(x)) = 0$ is n , then n is a factor of

(A) 4 (B) 6 (C) 3 (D) 5

[MATRIX TYPE]

Q.15 has four statements (A, B, C, D) given in **Column-I** and five statements (P, Q, R, S, T) given in **Column-II**. Any given statement in **Column-I** can have correct matching with one or more statement(s) given in **Column-II**.

Q.15 Let $\int x^2 \frac{d}{dx} \left(\frac{3x^2 + 1}{x^7 + 2x^5 + 2x^4 + x^3 + 2x^2 + 5x} \right) dx = f(x) - \tan^{-1}(g(x)) + c$,

where c is constant of integration and $f(0) = 0$, $g(1) = \frac{3}{2}$. Also, $h(g(x)) = x \forall x \in \mathbb{R}$

Column-I

Column-II

- | | |
|---|----------------|
| (A) If the value of $\tan^{-1}(g(0)) + \tan^{-1}\left(\frac{1}{2g(1)}\right) = \tan^{-1}(2g(p))$, then p is | (P) 0 |
| (B) If the number of solutions(s) of the equation $g(x) = h(x)$ is equal to $\frac{q}{\pi} \sin^{-1}(g(0))$, then q is | (Q) 1
(R) 3 |
| (C) If $h\left(\frac{31}{2}\right) = \frac{g(0)}{k}$, then k is | (S) 6 |
| (D) If $\int_{-3}^3 g(x) dx = 2g(n)$, then n is | (T) 7 |

[INTEGER TYPE]

Q.16 to Q.19 are "Integer Type" questions. (The answer to each of the questions are upto **4 digits**)

Q.16 If $A = \int_0^{101} [x] \sin \pi x \, dx$, then find the value of $\frac{A\pi}{2}$.

[Note: $[y]$ denotes greatest integer function less than or equal to y .]

Q.17 Let f be a twice differentiable function such that $f''(t) = \int_{12t}^{t^3+6t^2} \ln \left| \frac{x+8}{(t+2)^3 - x} \right| dx \quad \forall t \in \mathbb{R}$

and $f(1) = 2f(2) = 2$, then find the value of $\int_{-1}^1 f(x) \, dx$.

Q.18 If $\int_0^{1/2} \frac{\pi \, dx}{2(\cos^{-1} x)^2 \sqrt{1-x^2}} = \frac{p}{q}$, where p and q are relatively prime then find $(p+q)$.

Q.19 If $\int e^x \left(\frac{1-x^n}{1-x} \right) dx = e^x \cdot P(x) + C$ where $n \in \mathbb{N}$ and ' C ' is constant of integration and $P(0) = 620$ then find the value of n .

ANSWER KEY

Q.1	A	Q.2	B	Q.3	B	Q.4	C	Q.5	A
Q.6	C	Q.7	C	Q.8	B	Q.9	C	Q.10	D
Q.11	ABCD	Q.12	AB	Q.13	CD	Q.14	ABCD		
Q.15	(A) P; (B) S; (C) T; (D) Q			Q.16	5050	Q.17	6	Q.18	0003
Q.19	7								

